



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 2 2023/2024

SECV2113-HUMAN-COMPUTER INTERACTION

SECTION 1

PROJECT 2a

LECTURER: DR. IZYAN IZZATI BINTI KAMSANI

GROUP MEMBERS

STUDENT NAME	MATRIC NO
JOANNE CHING YIN XUAN	A23CS0227
EVELYN GOH YUAN QI	A23CS0222
LIM YU HAN	A23CS0241
TAN ZHI MING	A23CS0189
NG YU HIN	A23CS0148

Title: Gathering requirements -Task Analysis

Introduction

We will be testing our application on our users which are environmental enthusiasts and activists,online researchers and students on Google app .Three tasks will be searching locations,searching images and searching history.

Our aim is to obtain user experience feedback about how Google app features function properly, and evaluate how the app fulfils users' demands.From this we can make improvements or modifications to our project.

Three tasks have been chosen for observation including search locations, search images, and search history. To obtain a wide view of the app's functionality, these tasks will be completed by three users with various backgrounds. This will help us to figure out where we can improve and how effectively the app serves the users.

Task 1 – Search Location

a. online research  HCI project 2a User 1 Task 1-Search Location her

0. Search Location
1. Open Google app
2. Search for desired location
 - 2.1 Click “Search Bar”
 - 2.2 Key in the location
3. Click “Search”
 - 3.1 View Results
4. Open Maps
 - 4.1 Click “Directions”
5. Adjust the map details
 - 5.1 Choose mode of transport
 - 5.2 Choose type of visualisations of map
6. Click start

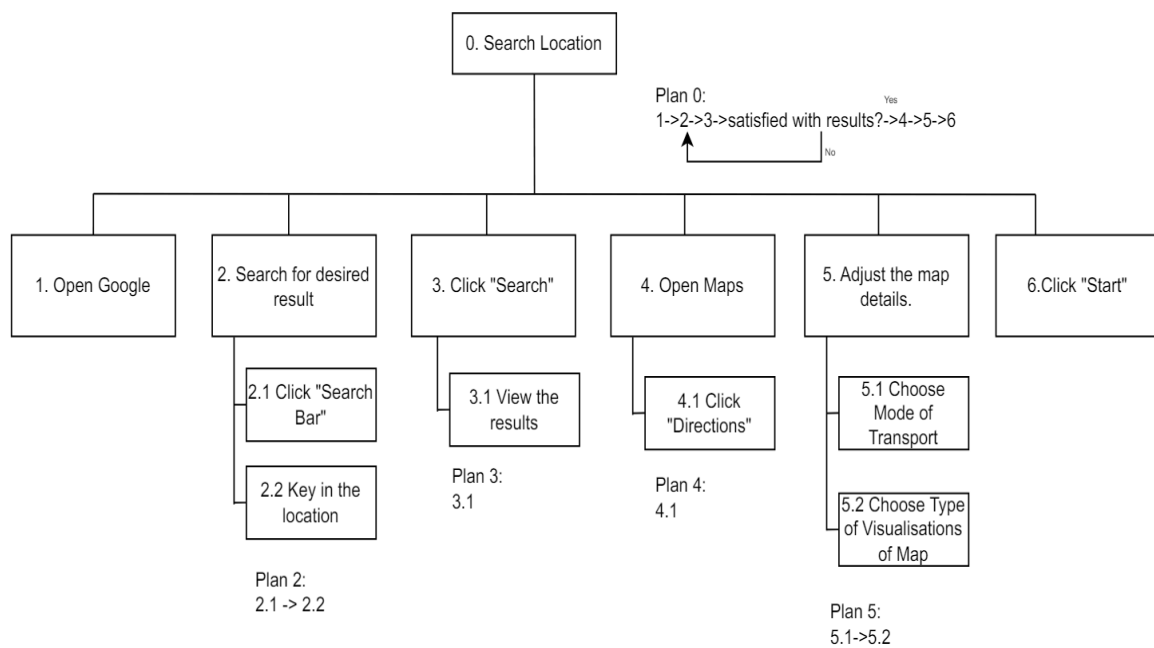
Plan 0 : Do 1 -> 2 -> 3 -> 4 -> 5 -> 6 , if 3 not satisfied, back to 2 again

Plan 2 : Do 2.1, then 2.2

Plan 3 : Do 3.1 to view & analyse the results

Plan 4 : Do 4.1 to show directions to destination

Plan 5 : Do 5.1, then 5.2 to have better visualisations



b. student

<https://youtu.be/AgTxnHck3I8>

0. Search location
1. Open Google app
2. Enter Search Query
 - 2.1 Click on search bar
 - 2.2 Type desired location
3. Execute the search
 - 3.1 Click "Search" button on keyboard
 - 3.2 Wait for search results to load
4. Review search result
 - 4.1 Check for the accuracy of information
5. View location
 - 5.1 Click maps
 - 5.2 Click directions

6. View transport
 - 6.1 Choose desired transport
7. Click start

Plan 0 : do 1 -> 2 -> 3 -> 4 -> 5 -> 6 -> 7, if 4 not satisfied, back to 2 again

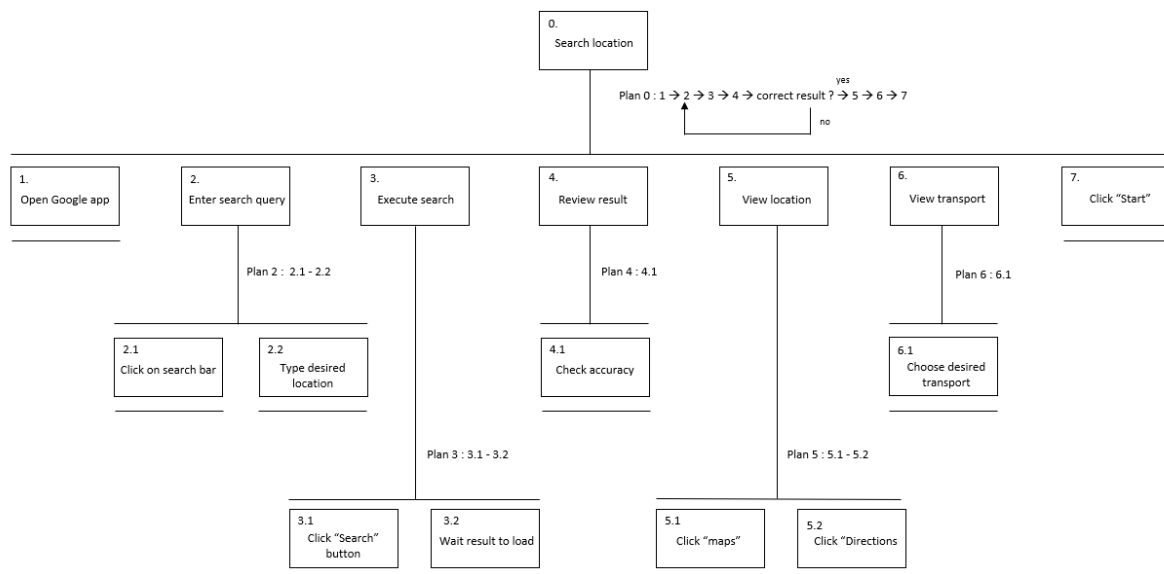
Plan 2 : do 2.1, then 2.2

Plan 3 : do 3.1, then 3.2

Plan 4 : do 4.1 to review result

Plan 5 : do 5.1, then 5.2 to view location

Plan 6 : do 6.1 to choose type of transportation



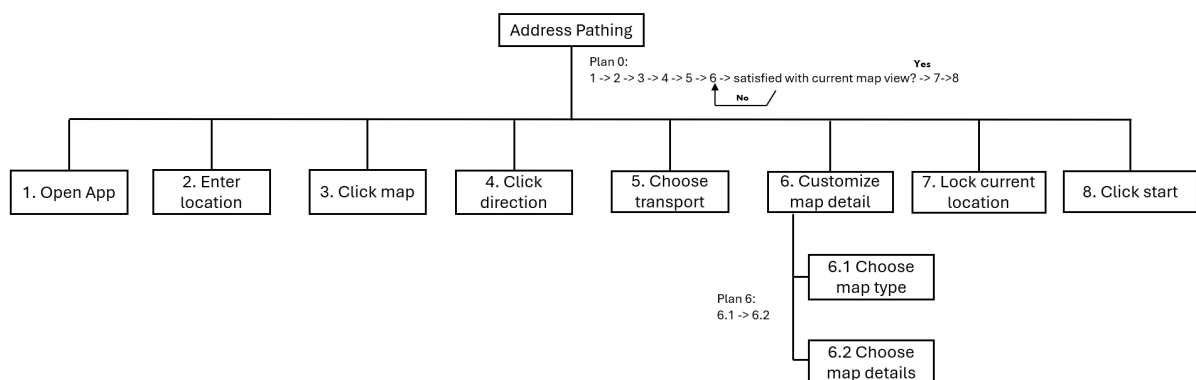
c. environment enthusiasts

<https://youtu.be/xq38XIQzxAc>

0. Search location path
1. Open the google app
2. Enter location in search bar
3. Click maps
4. Click directions
5. Choose type of transport
6. Customise map detail
 - 6.1 Choose Map type
 - 6.2 Choose Map detail
7. Click target current location icon
8. Click start

Plan 0: Do 1-> 2 -> 3 -> 4 -> 5 -> 6 -> if satisfied proceed 7, if not back to 6->7->8

Plan 6: Do 6.1->6.2



Findings From the HTAs for Task 1

- Due to the unclear icon, users are unaware of the function for selecting map types.
- And the option to switch between standard, satellite and terrain view is not easily distinguishable, which will confuse users who are not familiar with Google Maps or are new users.

- Thus, the icon for selecting map visualisations should be larger, and if possible, it should include a tutorial for using Google Maps to help new users learn and familiarize themselves with all of its functionalities.

Task 2 – Image Searching

a. online researcher

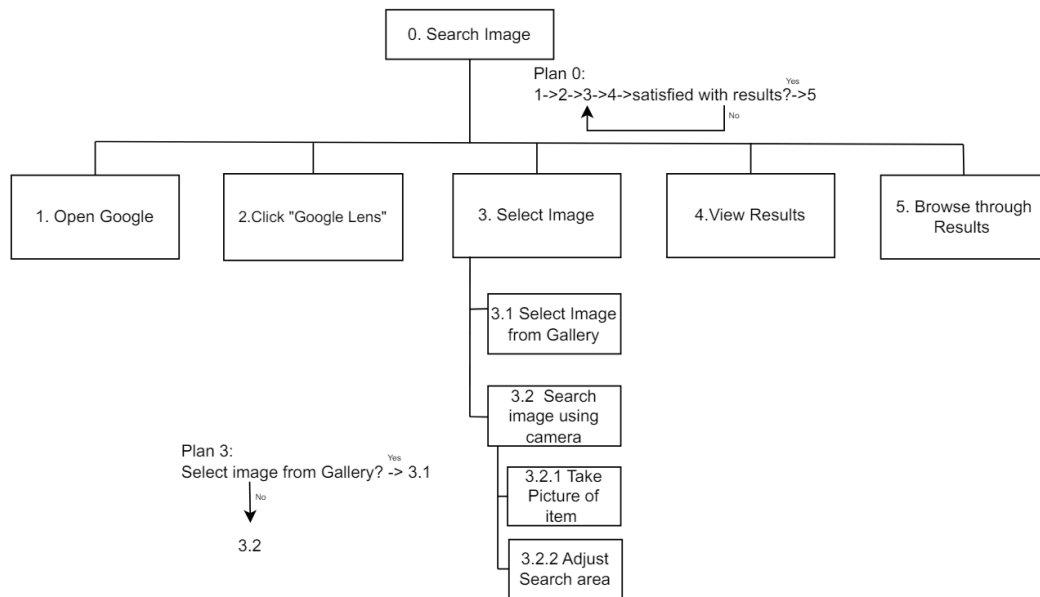
HCI project 2a User 1 Task 2-Search Image

0. Search Image
1. Open Google
2. Click “Google Lens”
3. Select image
 - 3.1 Select image from gallery
 - 3.2 Search image using camera
 - 3.2.1 Take picture of item
 - 3.2.2 Adjust search area
4. View Results
5. Browse through results

Plan 0: Do 1-> 2 -> 3 -> 4 ->5 , if 4 not satisfied,back to 3 again

Plan 3:Do 3.1 or 3.2 for methods to obtain image

Plan 3.2:Do 3.2.1, then 3.2.2 to take picture



b. student

<https://youtu.be/Zy1yIn-e4Sc>

0. Search image
1. Open Google app
2. Access Google Lens
 - 2.1 Tap camera icon
3. Select image
 - 3.1 select image from gallery
 - 3.1.1 browse through device's photo
 - 3.1.2 choose desired image
 - 3.2 capture image
 - 3.2.1 point device's camera at the object
 - 3.2.2 align image within the frame
4. Click search icon
5. Analyse result
 - 5.1 scan through search result

5.2 check for the accuracy of information

Plan 0 : do 1 -> 2 -> 3 -> 4 -> 5, if 5 not satisfied, back to 3 again

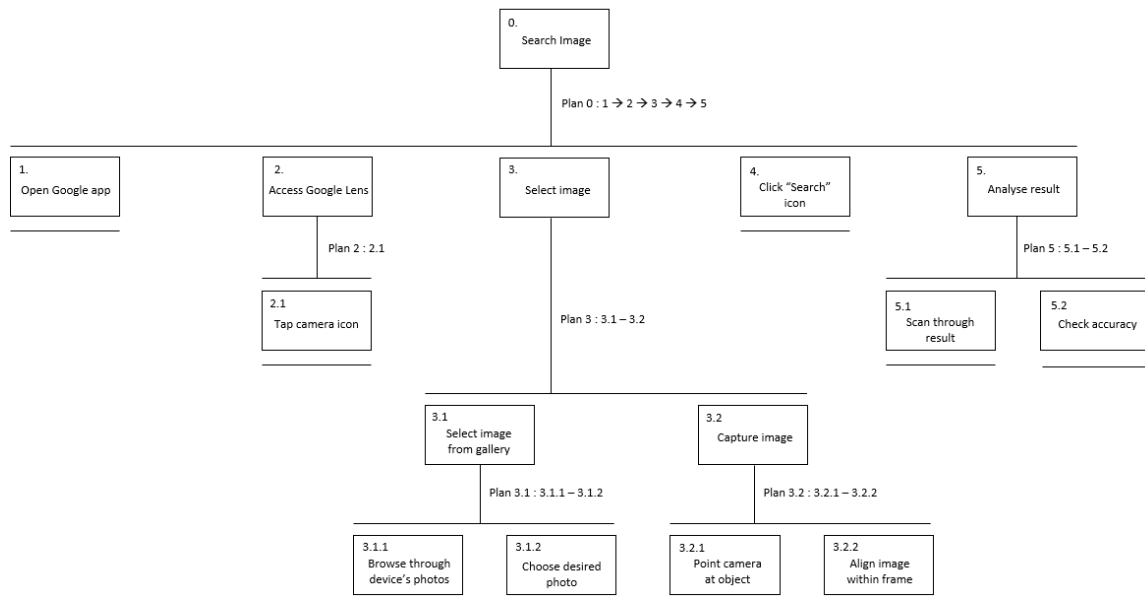
Plan 2 : do 2.1 to access Google Lens

Plan 3 : do 3.1 or 3.2 to select type to capture image

Plan 3.1 : do 3.1.1, then 3.1.2 to select image from gallery

Plan 3.2 : do 3.2.1, then 3.2.2 to capture image using camera

Plan 5 : do 5.1, then 5.2 to analyse result



c. environment enthusiasts

https://youtu.be/vb39pV5T_T0

0. Image searching

1. Open the google app

2. Click the button of image search

3. Select image

3.1 Select image from gallery

3.2 Search with camera

3.2.1 take photo for the item

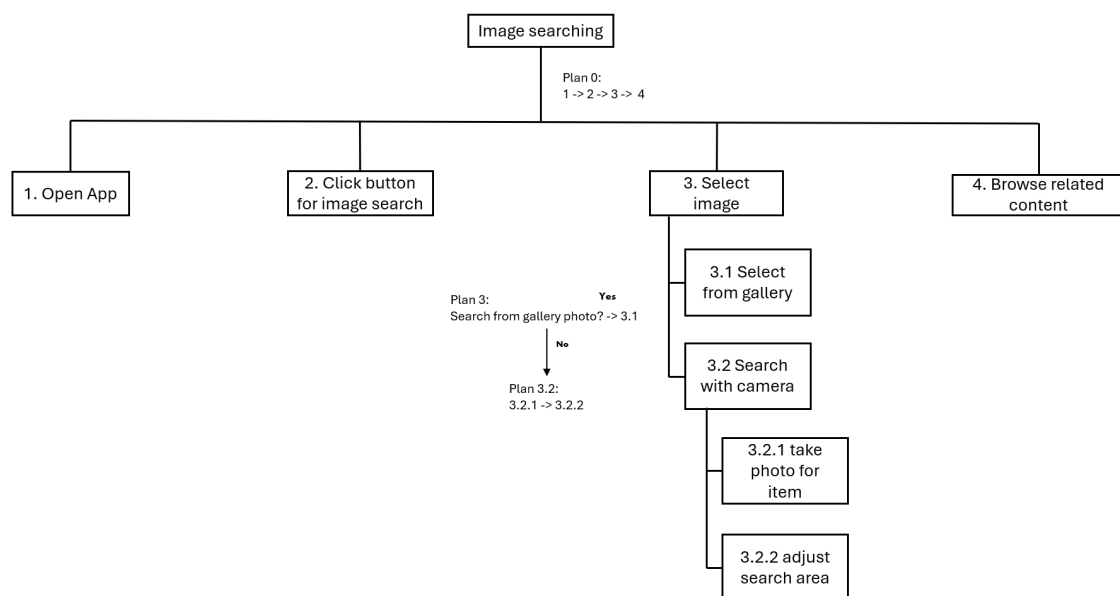
3.2.2 adjust search area

4. Browse related content

Plan 0: Do 1-> 2 -> 3 -> 4

Plan 3: If search from gallery photo proceed to 3.1 , if not proceed to 3.2

Plan 3.2 : Do 3.2.1 -> 3.2.2



Findings From the HTAs for Task 2

- Online researcher , student and environment enthusiast seems no problems were faced in image searching
- Operating the page is straightforward and simple
- Have clear ICONS to help users identify features
- The only thing that needs to be improved is the sharpness of the picture

Task 3 – History Searching

a. online researcher

▶ HCI project 2a User 1 Task 3-Search History

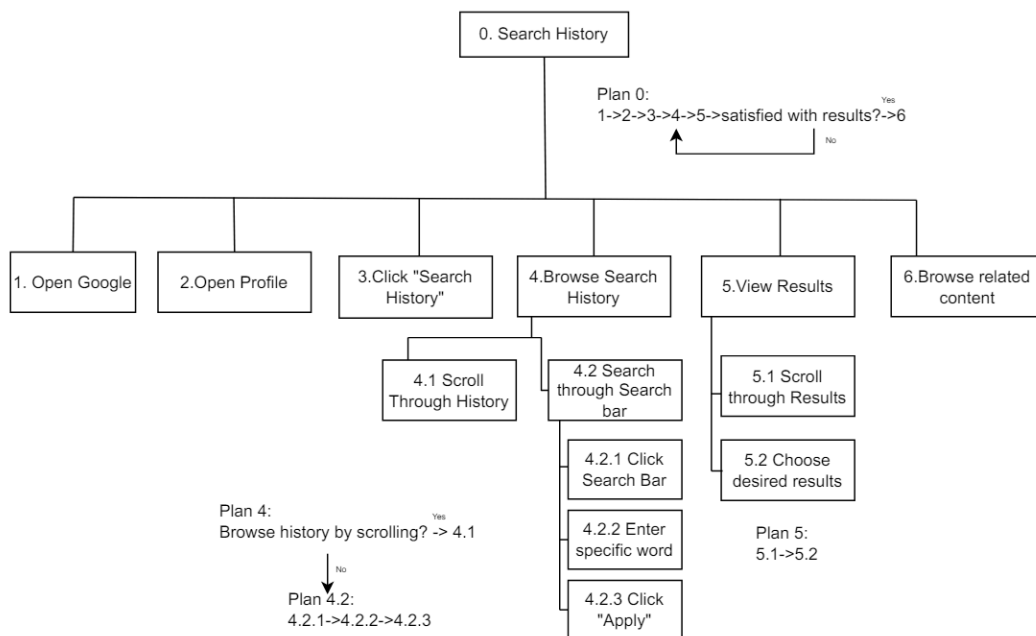
0. Search history
1. Open Google
2. Open profile
3. Click “Search History”
4. Browse Search History
 - 4.1 Scroll through history
 - 4.2 Search through Search Bar
 - 4.2.1 Click Search Bar
 - 4.2.2 Enter specific word
 - 4.2.3 Click “Apply”
5. View results
 - 5.1 Scroll through the result
 - 5.2 Choose desired results
6. Browse related content

Plan 0: Do 1->2->3->4->5->6 if 5, not satisfied back to 4 again.

Plan 4: Do 4.1 or 4.2 to browse search history

Plan 4.2 :Do 4.2.1, then 4.2.2 and finally 4.2.3.

Plan 5: Do 5.1, then 5.2.



b. student

https://youtu.be/Zu9Y5YCH_5E

0. Search history
1. Open Google app
2. Click G icon
3. Open profile
4. Click on "Search history"
5. Review search history
 - 5.1 look for desired search history
 - 5.2 manually search
 - 5.2.1 click magnifier icon
 - 5.2.2 enter search query
 - 5.2.3 click apply

6. Interpret search result
 - 6.1 scan through the result
 - 6.2 choose for desired title
 - 6.3 click desired search result
7. View browsing history
 - 7.1 Directed to a webpage
 - 7.2 Review the content

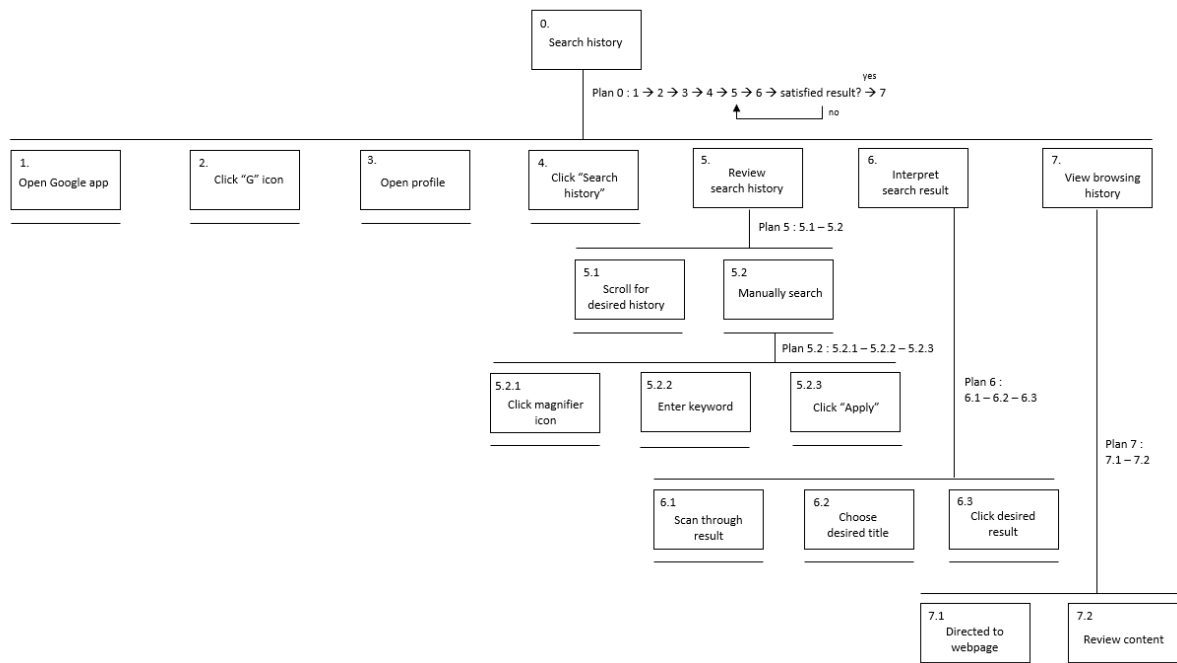
Plan 0 : do 1 -> 2 -> 3 -> 4 -> 5 -> 6 -> 7, if 6.1 not satisfied, back to 5 again

Plan 5 : do 5.1, then 5.2 to choose type to review history

Plan 5.2 : do 5.2.1, then 5.2.2, then 5.2.3 to search history manually

Plan 6 : do 6.1, then 6.2, then 6.3 to analyse search result

Plan 7 : do 7.1, then 7.2 to view history



c. environment enthusiasts

<https://youtu.be/HoerniWK9PM>

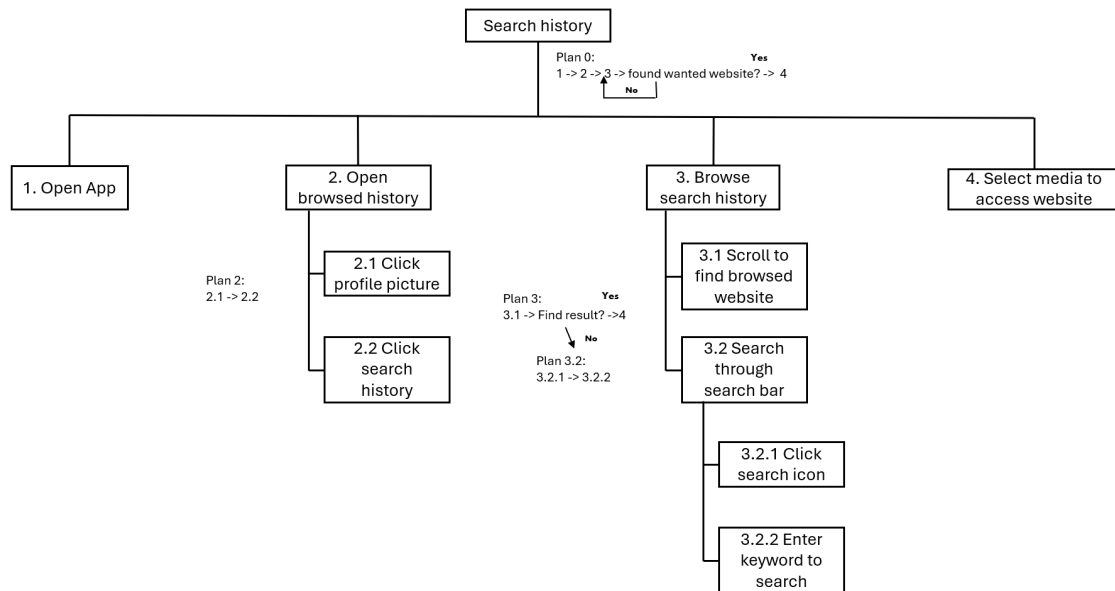
0. Search History
1. Open the google app
2. Open browsed history
 - 2.1 Click profile picture
 - 2.2 Click search history
3. Browse search history
 - 3.1 Scroll to find browsed website
 - 3.2 Search through search bar
 - 3.2.1 Click search icon
 - 3.2.2 Enter keyword to search
4. Select media to access website

Plan 0: Do 1-> 2 -> 3 -> 4

Plan 2 :Do 2.1->2.2

Plan 3: Do 3.1->if find result proceed 4 , if no proceed to 3.2

Plan 3.2: Do 3.2.1->3.2.2



Findings from the HTAs for Task 3

- Both the online researcher and student can't seem to find the visited website by scrolling, perhaps they both usually rely more on the internet to search for a vast variety of information that resulted in longer history lists, in contrast to the environment enthusiasts which rarely use the internet for detailed research.
- The search history is not categorised and grouped together which makes scrolling to search and visualising more difficult. For example, images are not categorised into one group but are displayed one by one.
- The search history should be categorised so it can be more efficient and convenient for displaying it.

Design requirements

Based on our research of existing application like Google, we have compiled a set of design requirements that will be incorporated into our redesigned version of Ecosia:

1. Implement a seamless and accurate address pathing feature that integrates with mapping services for precise location identification.
2. Develop comprehensive search history functionality that enables user to view,manage and delete their search history
3. Integrate image search capabilities by potentially using AI-driven technologies for visual search functionality similar to Google Lens

By analysing the users' thought processes, we have been able to clearly identify specific steps they take to complete designated tasks. This analysis has provided us with insights on how to improve our system's usability and efficiency.

Based on our research of existing application like Google, we have compiled a set of design requirements that will be incorporated into our redesigned version of Ecosia:

1. Implement a seamless and accurate address pathing feature that integrates with mapping services for precise location identification by improving the icon clarity and provide tutorial feature for first-time users. .
2. Integrate image search capabilities by potentially using AI-driven technologies for visual search functionality similar to Google Lens and enhanced the image quality that displayed in search results.
3. Develop comprehensive search history functionality that enables user to manage their search history and develop functionality to categorize search history into distinct groups.

By analysing the users' thought processes, we have been able to clearly identify specific steps they take to complete designated tasks. This analysis has provided us with insights on how to improve our system's usability and efficiency.